

Alerting in Pulse

In this page you can find a set of alerts that can be added to the monitoring solution in order to improve awareness over Pulse stability.



Note that these alerts are solemnly suggestions that might require new dashboards or alert adjustments based on HSBC specific environments.

Alert	Configuration	Description
Workflow Throughput	alert Level: Critical for alert: 1m frequency: 1m threshold value: 0.00001 (messages per second)	Measures the overall application workflow throughput per message type(normal, recovery, replica).
Workflow Errors	alert Level: High for alert: 10m frequency: 10s threshold value: 0 (events)	Measures the overall application workflow errors, considering all message types(normal, recovery and replica).
Events Dumped to Disk	alert Level: Critical for alert: 1m frequency: 1m threshold value: 0 (events)	Measures the overall application events spilled to disk. An event is spilled to disk when Pulse cannot write it on the event storage.
Auto Recovery Job	alert Level: Moderate for alert: 0s frequency: 5m threshold value: 0 (events)	Measures the overall number of events that are present in the in-memory queue to be written to the dump file in the failed folder, application and workflow.
Profile Fetches Latencies	alert Level: High for alert: 10m frequency: 1m threshold value: 0.4 (seconds)	Measures the overall max latency to fetch the profiles at percentile 99, per application, workflow and plan.
Pulse Application States	alert Level: Critical for alert: 12h frequency: 30s threshold value: 6 (STARTED)	Monitors the state of applications across all Pulse instances. If the application state is not STARTED (state < 6), an alert is triggered.
Pulse JVM GC Time	alert Level: High for alert: 10m frequency: 5m threshold value: 1 (seconds)	Measures maximum JVM GC duration. Long garbage collection pause durations indicate that the heap size is not adjusted for the actual usage.
App JVM GC Time	alert Level: High for alert: 10m frequency: 15m threshold value: 1 (seconds)	Measures maximum JVM GC duration for young and old gen per application and host, indicating heap size adjustment needs.
Pulse JVM Heap Usage	alert Level: Critical for alert: 5m frequency: 1m threshold value: 0.95 (percentage)	Measures JVM heap usage per Pulse instance, indicating high memory utilization.

Alert	Configuration	Description
App JVM Heap Usage	alert Level: Critical for alert: 5m frequency: 1m threshold value: 0.99 (percentage)	Measures JVM heap usage per application and host, indicating high memory utilization.
Pulse Threads Count	alert Level: Moderate for alert: 30m frequency: 10s threshold value: 2500 (threads)	Measures the current number of active threads in the JVM per Pulse instance.
App Threads Count	alert Level: Moderate for alert: 30m frequency: 10s threshold value: 3000 (threads)	Measures the current number of active threads in the JVM per application and host.
Pulse Heap Usage After GC	alert Level: High for alert: 15m frequency: 10s threshold value: 0.90 (percentage)	Measures the JVM heap usage after garbage collection per Pulse instance.
App Heap Usage After GC	alert Level: High for alert: 15m frequency: 10s threshold value: 0.9 (percentage)	Measures the JVM heap usage after garbage collection per application and host.
Pulse JVM File Descriptors Ratio	alert Level: High for alert: 10m frequency: 10s threshold value: 0.85 (percentage)	Measures the JVM file descriptors usage ratio per Pulse instance.
App JVM File Descriptors Ratio	alert Level: High for alert: 10m frequency: 10s threshold value: 0.85 (percentage)	Measures the JVM file descriptors usage ratio per application and host.
Pulse JVM Code Cache Usage	alert Level: Critical for alert: 10m frequency: 10s threshold value: 0.90 (percentage)	Measures the JVM code cache usage per Pulse instance.
App JVM Code Cache Usage	alert Level: Critical for alert: 10m frequency: 10s threshold value: 0.95 (percentage)	Measures the JVM code cache usage per application and host.
Messaging Healthcheck	alert Level: High for alert: 5m frequency: 2m threshold value: 1 (event)	Monitors the Pulse healthcheck for messaging communication with the messaging server.
Zookeeper Healthcheck	alert Level: High for alert: 5m frequency: 2m threshold value: 1 (event)	Monitors the Pulse healthcheck for Zookeeper communication.
Metadata DB Connection Pool	alert Level: Low for alert: 30m frequency: 1m	Monitors the metadata DB connection pool usage.

Alert	Configuration	Description
	threshold value: 0.9 (percentage)	
Metadata DB Connection Pool Per App	alert Level: Low for alert: 30m frequency: 1m threshold value: 0.9 (percentage)	Monitors the metadata connection pools usage per application.

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- alert Level : The severity of the alert (Low, Moderate, High, Critical).
- for alert : The duration for which the alert condition must persist before an alert is triggered.
- frequency : The interval at which the monitoring system checks the specified metrics or conditions.
- threshold value : A predefined limit or value that, when exceeded, triggers an alert.